Project	EventForce Management System (Salesforce)
Document	Project Design Phase
Team ID	SWTID-2026-1405 | SmartBridge x SkillWallet
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Project Design Phase

1. Solution architecture

EventForce follows the usual layered structure of a Salesforce application.

Layer	Components
Presentation	Event Planner Lightning app, object tabs, record pages, report and dashboard
Business logic	Validation rule on Client, Apex triggers on Event, record-triggered flow, approval process, scheduled batch job
Data	Event, Client, Venue, Vendor, Event Vendor and Feedback objects with relationships
Security	Profiles, role hierarchy, permission set, organization wide defaults and sharing rule

2. Data model

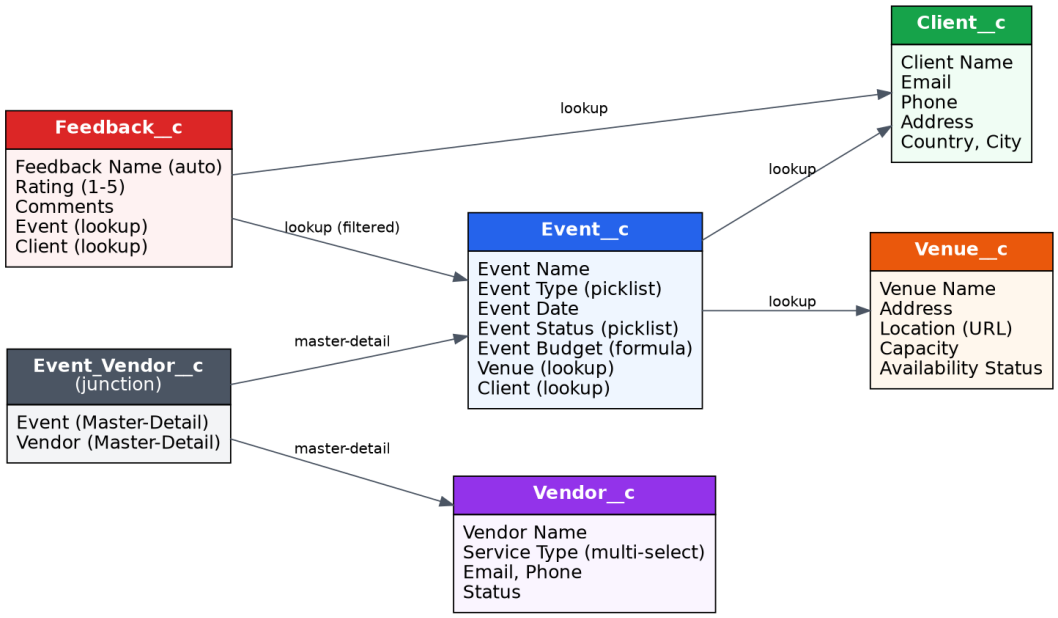


Figure 1: Object relationship diagram

Relationship	Type	Reason
Event to Venue	Lookup	A venue can host many events; an event may exist before a venue is final
Event to Client	Lookup	A client can book many events
Event Vendor to Event and Vendor	Master-Detail (two)	Many-to-many link: an event uses many vendors and a vendor serves many events

Relationship	Type	Reason
Feedback to Event	Lookup with filter	Only events of the selected client are offered
Feedback to Client	Lookup	Shows who gave the feedback

3. Process design

3.1 Automation overview

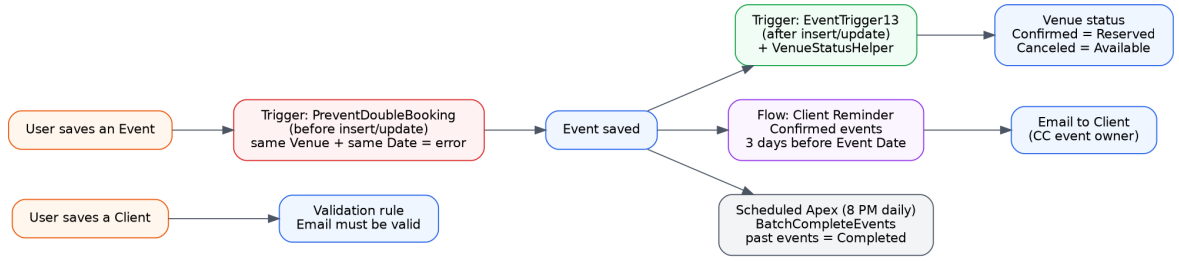


Figure 2: How validation, triggers, flow and scheduled Apex work together

3.2 Event cancellation approval

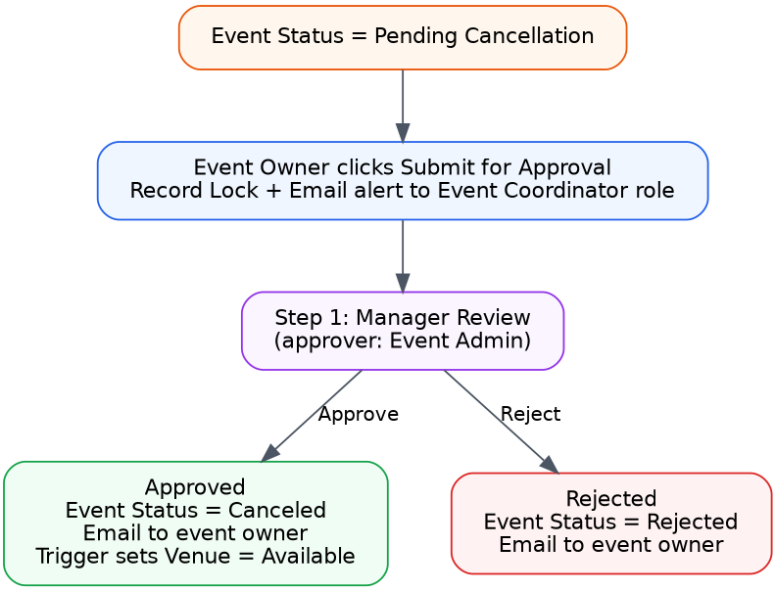


Figure 3: Cancellation approval process

4. User stories

As a	I want to	So that
Event Coordinator	create an event for a client at a venue	the booking is recorded and the budget is estimated
Event Coordinator	be stopped when I choose a venue that is already booked	no two clients get the same venue and date
Client	receive a reminder three days before my event	I am prepared for the day
Event Coordinator	request cancellation of an event	the admin can review it and the record is locked meanwhile

As a	I want to	So that
Event Admin	approve or reject the request	the event status and venue availability update automatically
Vendor Manager	maintain vendor details	the coordinators always see current vendor information
Event Admin	see upcoming events grouped by month	I can plan resources and budget

5. Security design

- Role hierarchy: Event Admin on top with Event Coordinator, Vendor Manager and Client below it.
- Four custom profiles give object level access; the Feedback Manager permission set gives extra Feedback access to the coordinator.
- Event is Private in the organization wide defaults. A sharing rule shares events owned by the Event Coordinator role with the Vendor Manager role as read only.
- Only the Event Admin approves cancellation requests.

6. Design decisions

- A junction object with two master-detail relationships is the standard way to model many-to-many in Salesforce.
- Venue update logic is placed in a helper class so the trigger stays small and the code can be reused.
- The double booking check runs before save, so invalid records never reach the database.
- Reminders use a scheduled path in a flow, which avoids custom code for a time based action.